Naram Mhaisen

Curriculum Vitae

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Research interests: Online learning; learning with predictions; non-stochastic control; networked systems.

Google Scholar | ResearchGate | GitHub | LinkedIn

Selected Research Contributions

1. Online learning with imperfect predictions. Developed optimistic algorithms for continuous, discrete, and stateful sequential decision problems. *ACM SIGMETRICS 2022, IEEE CDC 2024, ICML 2025.*

2. Online learning with movement costs. Introduced a partially lazy gradient method that attains optimal dynamic regret while preserving decision stability. *AISTATS 2026 and ongoing work.*

3. Applications of online learning. Applied to network caching, routing, slicing, and navigation problems. *IEEE TMC 2023, IEEE TNSM 2025, ESWA 2025, Foundation and Trends in Networking 2026.*

Education

04/2021–11/2025 **Ph.D. in Computer Science**, *Delft University of Technology*, Netherlands.
- Thesis: "Optimistic Learning with Applications to Caching Networks"
- Supervisors: George Iosifidis, Koen Langendoen.
- Examiners: T. van Erven, G. Dán, P. Grosso, M. Spaan, T. Spyropoulos.

09/2018–06/2020 **Master of Science in Computing**, *Qatar University*, Qatar, *GPA: 3.88/4.00*
- Thesis: "Novel Techniques for Blockchain-enabled IoT leveraging Reinforcement Learning"
- Supervisor: Mohsen Guizani.
- "Distinguished Research" Award for obtaining >3.8 GPA & publishing the findings.

09/2013–08/2017 **Bachelor of Science in Computer Engineering**, *Qatar University*, Qatar, *GPA: 4.00/4.00*
- "Order of Excellence" Award for obtaining >3.8 GPA.
- "Honors" Award for completing Qatar University Honors Program.

Experience

11/2025–Present **Postdoctoral Researcher**, *Delft University of Technology*, Netherlands
- Developing online learning and optimization methods within ORIGAMI, a Horizon Europe project, for adaptive resource allocation in distributed communication and computing networks, with emphasis on nonstationarity, prediction-assisted decisions, and reconfiguration costs.

07/2020–04/2021 **Research Assistant**, *Qatar University*, Qatar
- R&D in a "High Impact" project on edge computing and secure medical-data exchange, with emphasis on scalable and privacy-aware software infrastructure.

10/2018–07/2020 **Graduate Research/Teaching Assistant**, *Qatar University*, Qatar
- Developed reinforcement-learning methods for adaptive resource allocation and decision-making in heterogeneous IoT and communication systems.
- Contributed to a Cyber Range challenge team in collaboration with Thales.

09/2017–08/2018 **Research Assistant**, *Qatar University*, Qatar
- R&D at "National Priority Project": "Efficiently reliable and privacy-preserving cloud storage".

07/2017– 09/2017 **Software Engineer**, *Qatar Mobility Innovation Center (QMIC)*, Qatar
- Hardware/Software design and implementation of a prepaid energy embedded system based on Ethereum Blockchain and IoT technologies.

Awards

2022 "Research Fellowship" Award, NEC Laboratories-Europe

2020 "Best Demo Award" IEEE Inter. Conference for Informatics, IoT, and Enabling Tech

2017 "Microsoft Imagine Cup" Winner (Qatar round & MEA Round, Worldwide Finalist)

2017 "Best Bachelor Senior Project" Award, by Qatar National Research Fund (QNRF).

2017 Qatar University Awards for "Non-Academic Activities" and "Voluntary Activities"

2016 & 2017 First Place in The GCC Robotics Challenge - Qatar Round

Talks

Tutorials

2024 "Optimistic Learning for Resource Management in Communication Systems" (Joint with G. Iosifidis) at ACM SIGMETRICS, Venice, Italy.

Seminars

2024 "Optimistic Learning", ELLIS Network Seminars, TU Delft.

2023 "Adversarial Online Learning" at the Department of Computer Science. Qatar University.

Conferences

2026	*AISTATS*	Tangier, Morocco. (Poster presentation)
2025	*ICML*	Vancouver, Canada. (Poster presentation)
2024	*IEEE CDC*	Milan, Italy. (Oral presentation)
2024	*L4DC*	University of Oxford, UK. (Poster presentation)
2022	*IFIP Networking*	University of Catania, Sicily, Italy. (Oral presentation)

Funding, Service, and Community

Funding: Contributed to the proposal development of ORIGAMI, a Horizon Europe project on learning-based resource allocation in communication networks.

Reviewing: Reviewer for ICML, NeurIPS, AISTATS, IEEE INFOCOM, IEEE Trans. on Mobile Computing, IEEE Trans. on Network Science and Engineering, and IEEE/ACM Trans. on Networking.

Teaching and Supervision

2024 **CSE 3000 Research Project**, *TU Delft*
- Supervised three bachelor research projects and participated in the grading:
  - *Machine Learning Algorithms for Caching Systems* (R. Vadastreanu)
  - *Meta-learning the Best Caching Expert* (M. de Vries)
  - *Optimistic Discrete Caching with Switching Costs* (L. Tosa)

2022 **CSE 3000 Research Project**, *TU Delft*
- Supervised a bachelor research project and participated in the grading:
  - *Integral Caching using Online Mirror Descent* (Q. Oschatz)

2022 **IN 4390: Quantitative Evaluation of Embedded Systems**, *TU Delft*
- Conducted lab sessions for practical assignments and additional exercises.
- Graded assignments.

Publication List

Conference Papers

[C14] **N. Mhaisen**, G. Iosifidis. “Partially Lazy Gradient Descent for Smoothed Online Learning”. In: *Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS)*. 2026.

[C13] **N. Mhaisen**, G. Iosifidis. “On the Dynamic Regret of Following the Regularized Leader: Optimism with History Pruning”. In: *Forty-second International Conference on Machine Learning (ICML)*. 2025.

[C12] N. Khial, **N. Mhaisen**, M. Mabrok, A. Mohamed. “An Online Learning Framework for UAV Target Search Missions in Non-Stationary Environments”. In: *2024 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*. 2024.

[C11] **N. Mhaisen**, G. Iosifidis. “Adaptive online non-stochastic control”. In: *Annual Learning for Dynamics & Control Conference (L4DC)*. PMLR. 2024.

[C10] **N. Mhaisen**, G. Iosifidis. “Optimistic Online Non-stochastic Control via FTRL”. In: *Conference on Decision and Control (CDC)*. 2024.

[C9] **N. Mhaisen**, A. Sinha, G. Paschos, G. Iosifidis. “Optimistic No-regret Algorithms for Discrete Caching”. In: *Abstract Proceedings of the ACM SIGMETRICS Conference on Measurement and Modeling of Computer Systems*. 2023.

[C8] J. Zheng, K. Li, **N. Mhaisen**, W. Ni, E. Tovar, M. Guizani. “Federated learning for online resource allocation in mobile edge computing: A deep reinforcement learning approach”. In: *2023 IEEE Wireless Communications and Networking Conference (WCNC)*. 2023.

[C7] **N. Mhaisen**, G. Iosifidis, D. Leith. “Online caching with optimistic learning”. In: *IFIP Networking Conference (IFIP Networking)*. 2022.

[C6] **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. “Rational Contracts: Data-driven Service Provisioning in Blockchain-powered Systems”. In: *ICC-IEEE International Conference on Communications*. 2021.

[C5] **N. Mhaisen**, N. Fetais, A. Massoud. “Real-time scheduling for electric vehicles charging/discharging using reinforcement learning”. In: *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT)*. 2020.

[C4] **N. Mhaisen**, O. Abazeed, Y. Al-Hariri, N. Nawaz, A. Amira. “A reconfigurable multipurpose system on chip platform for metal detection”. In: *IEEE 10th GCC Conference & Exhibition (GCC)*. 2019.

[C3] O. Al Rshid Abazeed, **N. Mhaisen**, Y. Al-Hariri, N. Nawaz, A. Amira. “A Reconfigurable Multipurpose SoC Mobile Platform for metal detection”. In: *Qatar Foundation Annual Research Conference Proceedings*. 2018.

[C2] **N. Mhaisen**, O. Abazeed, Y. Al Hariri, A. Alsalemi, O. Halabi. “Self-powered IoT-enabled water monitoring system”. In: *International Conference on Computer and Applications (ICCA)*. 2018.

[C1] **N. Mhaisen**, M. Punkar, Y. Wang, Y. Desmedt, Q. Malluhi. “Almost BPXOR Coding Technique for Tolerating Three Disk Failures in RAID7 Architectures”. In: *Qatar Foundation Annual Research Conference Proceedings*. 2018.

Journal Articles

[J15] M. Helmy, A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad. “Slicing for AI: An Online Learning Framework for Network Slicing Supporting AI Services”. In: *IEEE Transactions on Network and Service Management* (2025).

[J14] N. Khial, M. S. Allahham, **N. Mhaisen**, L. Ismail, M. Mabrok, A. Mohamed. “DRONE-RL: Dynamic Reinforcement learning for Online Navigation of UAVs in Evolving environments”. In: *Knowledge-Based Systems* (2025).

[J13] N. Khial, **N. Mhaisen**, M. Mabrok, A. Mohamed. “An online learning framework for UAV search mission in adversarial environments”. In: *Expert Systems with Applications* 267 (2025).

[J12] A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. “Reinforcement learning for intelligent healthcare systems: A review of challenges, applications, and open research issues”. In: *IEEE Internet of Things Journal* 10.24 (2023).

[J11] **N. Mhaisen**, G. Iosifidis, D. Leith. “Online caching with no regret: Optimistic learning via recommendations”. In: *IEEE Transactions on Mobile Computing* 23.5 (2023).

[J10] **N. Mhaisen**, A. Sinha, G. Paschos, G. Iosifidis. “Optimistic no-regret algorithms for discrete caching”. In: *Proceedings of the ACM on Measurement and Analysis of Computing Systems (SIGMETRICS)* 6.3 (2022).

[J9] A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani, Z. Dawy, W. Nasreddine. “Communication-efficient hierarchical federated learning for IoT heterogeneous systems with imbalanced data”. In: *Future Generation Computer Systems* 128 (2022).

[J8] M. S. Allahham, A. A. Abdellatif, **N. Mhaisen**, A. Mohamed, A. Erbad, M. Guizani. “Multi-agent reinforcement learning for network selection and resource allocation in heterogeneous multi-RAT networks”. In: *IEEE Transactions on Cognitive Communications and Networking* 8.2 (2022).

[J7] E. Baccour, **N. Mhaisen**, A. A. Abdellatif, A. Erbad, A. Mohamed, M. Hamdi, M. Guizani. "Pervasive AI for IoT applications: A Survey on Resource-efficient Distributed Artificial Intelligence". In: *IEEE Communications Surveys & Tutorials* (2022).

[J6] J. Zheng, K. Li, **N. Mhaisen**, W. Ni, E. Tovar, M. Guizani. "Exploring deep-reinforcement-learning-assisted federated learning for online resource allocation in privacy-preserving EdgeIoT". In: *IEEE Internet of Things Journal* 9.21 (2022).

[J5] **N. Mhaisen**, A. A. Abdellatif, A. Mohamed, A. Erbad, M. Guizani. "Optimal user-edge assignment in hierarchical federated learning based on statistical properties and network topology constraints". In: *IEEE Transactions on Network Science and Engineering* 9.1 (2021).

[J4] **N. Mhaisen**, M. S. Allahham, A. Mohamed, A. Erbad, M. Guizani. "On designing smart agents for service provisioning in blockchain-powered systems". In: *IEEE Transactions on Network Science and Engineering* 9.2 (2021).

[J3] **N. Mhaisen**, N. Fetais, A. Erbad, A. Mohamed, M. Guizani. "To chain or not to chain: A reinforcement learning approach for blockchain-enabled IoT monitoring applications". In: *Future Generation Computer Systems* 111 (2020).

[J2] **N. Mhaisen**, Q. M. Malluhi. "Data consistency in multi-cloud storage systems with passive servers and non-communicating clients". In: *IEEE Access* 8 (2020).

[J1] **N. Mhaisen**, N. Fetais, A. Massoud. "Secure smart contract-enabled control of battery energy storage systems against cyber-attacks". In: *Alexandria Engineering Journal* 58.4 (2019).

Patents

[P2] Q. Malluhi, **N. Mhaisen**. "Data storage methods and systems". US Patent 11,748,197. Sept. 2023.

[P1] Q. Malluhi, **N. Mhaisen**. "Data storage methods and systems". US Patent 11,531,647. Dec. 2022.

Monograph & Preprints

[A2] G. Iosifidis, **N. Mhaisen**, D. Leith "Optimistic Learning for Communication Networks" arXiv 2504.03499. *Monograph accepted in Foundation and Trends (FnT) in Networking, expected publication Q4 2026.*

[A1] **N. Mhaisen**, G. Iosifidis "Dynamic Regret for Online Convex Optimization with Indicator Switching Costs". *Submitted to NeurIPS 2026.*

References

Dr. Georgios (George) Iosifidis, Associate Professor
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